

Angles in Quadrilaterals — Practice

Section A is interactive (including a figure question you can self-check). Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 • MCQ

[1]

The sum of the interior angles of a quadrilateral is:

- (a) 180° (b) 270° (c) 360° (d) 720°

Q2 • MCQ

[1]

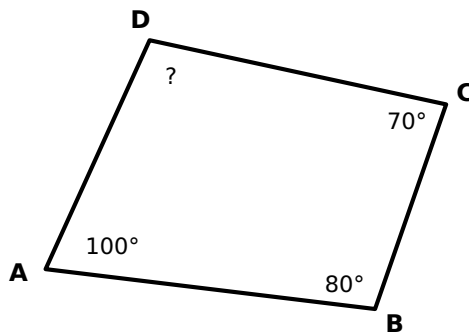
The number of diagonals in a polygon of n sides is:

- (a) n (b) $n - 3$ (c) $n(n - 3)/2$ (d) $n(n + 1)/2$

Q3 • FIGURE • FILL IN THE BLANK

[1]

Find $\angle D$ in quadrilateral ABCD shown below (answer in degrees).



Q4 • MCQ

[1]

Each exterior angle of a regular polygon is 45° . The number of sides is:

- (a) 6 (b) 8 (c) 9 (d) 10

Q5 · TRUE / FALSE

[1]

The sum of the exterior angles of any polygon is 360° .

True False

Q6 · ASSERTION-REASON

[1]

Assertion (A): The sum of the interior angles of a quadrilateral is 360° .

Reason (R): A quadrilateral can be divided into two triangles.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7

[2]

The angles of a quadrilateral are in the ratio 1 : 2 : 3 : 4. Find the measure of each angle.

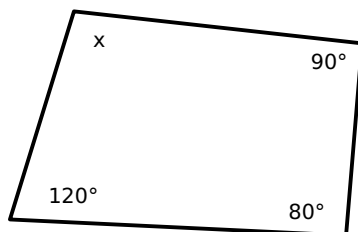
Q8

[2]

Find the number of sides of a regular polygon whose exterior angle measures 30° .

Q9 · FIGURE**[2]**

Find the value of x in the quadrilateral below.



Long answer (3 marks each)**Q10****[3]**

The four angles of a quadrilateral are x , $2x$, 80° and 100° . Find x and hence all four angles.

Q11**[3]**

A regular polygon has 9 sides. Find (a) each exterior angle, (b) each interior angle, (c) the sum of all interior angles.

Q12

[3]

How many diagonals does an octagon (8 sides) have? Show the formula and your working.

Total: 21 marks · [Check your work against the answer key.](#)